Two Special Signals for LTI Systems

$$x(t) = e^{st} \longrightarrow \boxed{h(t)} \longrightarrow$$

$$\begin{aligned} y(t) &= \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st} \\ &= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}} \end{aligned}$$

$$x[n] = z^n \longrightarrow \boxed{h[n]} \longrightarrow$$

$$\begin{aligned} y[n] &= \sum_{m=-\infty}^{\infty} h[m] z^{n-m} \\ &= \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m} \right] z^n \\ &= \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}} \end{aligned}$$

System Functions *H*(*s*) or *H*(*z*)

CT:

$$x(t) \xrightarrow{e^{st}} \boxed{h(t)} \xrightarrow{H(s)e^{st}} y(t)$$

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}\,dt$$

$$x(t) = \sum a_k e^{s_k t} \longrightarrow y(t) = \sum H(s_k)a_k e^{s_k t}$$

DT:

$$x[n] \xrightarrow{z^n} \boxed{h[n]} \xrightarrow{H(z)z^n} y[n]$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

$$x[n] = \sum a_k z_k^n \longrightarrow y[n] = \sum H(z_k)a_k z_k^n$$

Fourier and Beyond

Observation: if one signal can be written as the linear combination of $e^{st} or\ z^n$, we need NOT to calculate the convolution for the LTI output.

When $s = j\omega, z = e^{j\omega}$

$$\Rightarrow\ e^{j\omega t}, e^{j\omega n}:\ Fourier\ Series$$

When s or z is general complex number

$$\Rightarrow Laplace\ Transform\ \&\ Z\ Transform$$

A "Special" Class of Periodic Signals

CT Fourier Series: one periodic CT signal with period T can be written as

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

where $\omega_0 = 2\pi/T$.

$\{a_k\}$: Fourier series coefficients, which represent the strength of the component $e^{jk\omega_0 t}$.

$e^{jk\omega_0 t}$ is a signal with pure frequency $k\omega_0 \Rightarrow x(t)$ is a periodic signal with period T, it consists of components with different frequencies $k\omega_0$ and different weights.

CT Fourier Series Pair

CT Fourier Series Pair $\qquad (\omega_o = 2\pi / T)$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t} \qquad \text{(Synthesis equation)}$$

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_o t} dt \qquad \text{(Analysis equation)}$$

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left.\begin{array}{l} x(t) \\ y(t) \end{array}\right\}$ Periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$	a_k b_k
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t}x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a^*_{-k}
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk\frac{2\pi}{T}a_k$
Integration		$\int_{-\infty}^{t} x(t)\,dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a^*_{-k} \\ \Re e\{a_k\} = \Re e\{a_{-k}\} \\ \Im m\{a_k\} = -\Im m\{a_{-k}\} \\ \lvert a_k\rvert = \lvert a_{-k}\rvert \\ \sphericalangle a_k = -\sphericalangle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \mathcal{E}v\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}d\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\Re e\{a_k\}$ $j\Im m\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

1 3.8

3.8. Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is real and odd.
2. $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients a_k.
3. $a_k = 0$ for $|k| > 1$.
4. $\frac{1}{2}\int_0^2 |x(t)|^2\, dt = 1$.

Specify two different signals that satisfy these conditions.

1 3.8

3.8. Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is real and odd.
2. $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients a_k.
3. $a_k = 0$ for $|k| > 1$.
4. $\frac{1}{2}\int_0^2 |x(t)|^2\,dt = 1$.

Specify two different signals that satisfy these conditions.

1. a_k are purely imaginary and odd

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$$

$$a_k^* = \frac{1}{T}\int_T x^*(t)e^{jk\omega_0 t}dt \qquad (e^{i\varphi} = \cos\varphi + i\sin\varphi)$$

$$a_{-k}^* = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = a_k \Rightarrow \begin{array}{l} \mathrm{Re}\{a_k\} = \mathrm{Re}\{a_{-k}\} \\ \mathrm{Im}\{a_k\} = -\mathrm{Im}\{a_{-k}\} \end{array}$$

$$x(-t) = -x(t)$$

$$x(-t) \to a_{-k}$$

$$-x(t) \to -a_k$$

$$a_{-k} = -a_k$$

1 3.8

3.8. Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is real and odd.

2. $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients a_k.

3. $a_k = 0$ for $|k| > 1$.

4. $\frac{1}{2}\int_0^2 |x(t)|^2 \, dt = 1$.

Specify two different signals that satisfy these conditions.

1. a_k are purely imaginary and odd. $a_0 = 0$

2. $\omega_0 = \pi$

3. Only a_1 and a_{-1} have values.

4. $|a_1|^2 + |a_{-1}|^2 = 1$

Parseval's relation:

$$\frac{1}{T}\int_{<T>} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

1 3.8

3.8. Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is real and odd.
2. $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients a_k.
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Specify two different signals that satisfy these conditions.

1. a_k are purely imaginary and odd. $a_0 = 0$

2. $\omega_0 = \pi$

3. Only a_1 and a_{-1} have values.

4. $|a_1|^2 + |a_{-1}|^2 = 1$

$$a_1 = -a_{-1} = \frac{1}{\sqrt{2}j} \text{ or } a_1 = -a_{-1} = -\frac{1}{\sqrt{2}j}$$

$$x_1(t) = \frac{1}{\sqrt{2}j} e^{j(2\pi/2)t} - \frac{1}{\sqrt{2}j} e^{-j(2\pi/2)t} = -\sqrt{2}\sin(\pi t) \text{ or}$$

$$x_1(t) = -\frac{1}{\sqrt{2}j} e^{j(2\pi/2)t} + \frac{1}{\sqrt{2}j} e^{-j(2\pi/2)t} = \sqrt{2}\sin(\pi t)$$

3.34

3.34. Consider a continuous-time LTI system with impulse response

$$h(t) = e^{-4|t|}.$$

Find the Fourier series representation of the output $y(t)$ for each of the following inputs:

(a) $x(t) = \sum_{n=-\infty}^{+\infty} \delta(t-n)$

(b) $x(t) = \sum_{n=-\infty}^{+\infty} (-1)^n \delta(t-n)$

(c) $x(t)$ is the periodic wave depicted in Figure P3.34.

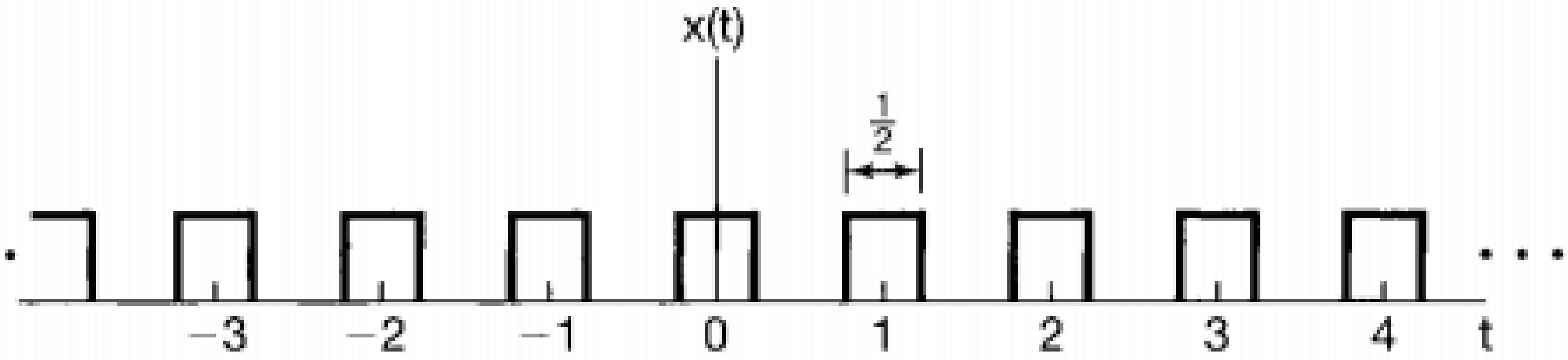

Figure P3.34

Two Special Signals for LTI Systems

$$x(t) = e^{st} \longrightarrow \boxed{h(t)} \longrightarrow$$

$$y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau$$

$$= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau\right] e^{st}$$

$$= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}}$$

$$x[n] = z^n \longrightarrow \boxed{h[n]} \longrightarrow$$

$$y[n] = \sum_{m=-\infty}^{\infty} h[m] z^{n-m}$$

$$= \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m}\right] z^n$$

$$= \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}}$$

System Functions *H*(*s*) or *H*(*z*)

CT:

$$x(t) \xrightarrow{e^{st}} \boxed{h(t)} \xrightarrow{H(s)e^{st}} y(t)$$

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}dt$$

$$x(t) = \sum a_k e^{s_k t} \longrightarrow y(t) = \sum H(s_k)a_k e^{s_k t}$$

DT:

$$x[n] \xrightarrow{z^{n}} \boxed{h[n]} \xrightarrow{H(z)z^{n}} y[n]$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

$$x[n] = \sum a_k z_k^n \longrightarrow y[n] = \sum H(z_k)a_k z_k^n$$

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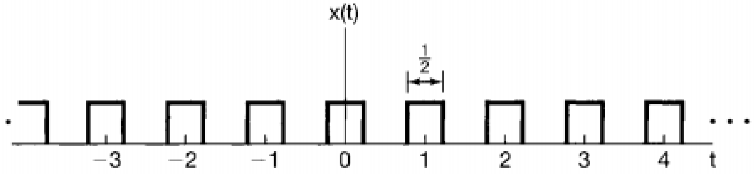

Figure P3.34

$$H(j\omega) = \int_{-\infty}^{\infty} e^{-4|t|} e^{-j\omega t} dt = \frac{1}{4+j\omega} + \frac{1}{4-j\omega}$$

$$(a)\ T=1,\ \omega_0 = 2\pi,\ a_k = 1$$

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{1}\int_{-1/2}^{1/2} \sum_{n=-\infty}^{+\infty} \delta(t-n) e^{-jk\omega_0 t} dt$$

$$= \int_{-1/2}^{1/2} \delta(t) e^{-jk\omega_0 t} dt$$

$$= e^{-jk\omega_0 0}$$

$$= 1$$

$$b_k = a_k H(j\omega_0 k) = \frac{1}{4+j2\pi k} + \frac{1}{4-j2\pi k}$$

$$(b)\ T=2,\ \omega_0 = \pi,\ a_k = \begin{cases} 0 &, \text{k is even} \\ 1 &, \text{k is odd} \end{cases}$$

$$b_k = a_k H(jk\omega_0) = \begin{cases} 0 &, \text{k is even} \\ \dfrac{1}{4+jk\pi} + \dfrac{1}{4-jk\pi} &, \text{k is odd} \end{cases}$$

1 3.34

3.34. Consider a continuous-time LTI system with impulse response

$$h(t) = e^{-4|t|}.$$

Find the Fourier series representation of the output $y(t)$ for each of the following inputs:

(a) $x(t) = \sum_{n=-\infty}^{+\infty} \delta(t-n)$

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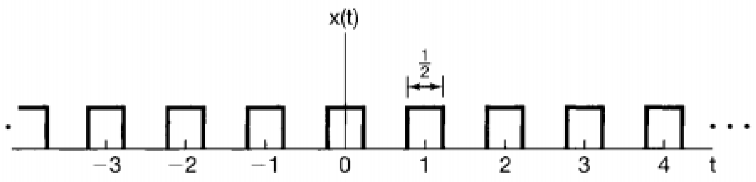

Figure P3.34

$$(c)\ T=1,\ \omega_0=2\pi,\ a_k=\begin{cases} 1/2 & ,\ k=0 \\ 0 & ,\ k \text{ is even},\ k\neq 0 \\ \dfrac{\sin(\pi k/2)}{\pi k} & ,\ k \text{ is odd} \end{cases}$$

$$b_k = a_k H(jk\omega_0) = \begin{cases} 1/4, & k=0 \\ 0, & k \text{ is even},\ k\neq 0 \\ \dfrac{\sin(\pi k/2)}{\pi k}\left[\dfrac{1}{4+j2\pi k}+\dfrac{1}{4-j2\pi k}\right], & k \text{ is odd} \end{cases}$$

1 3.40

3.40. Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k. Derive the Fourier series coefficients of each of the following signals in terms of a_k:

(a) $x(t - t_0) + x(t + t_0)$

(b) $\mathcal{E}v\{x(t)\}$

(c) $\mathcal{R}e\{x(t)\}$

(d) $\frac{d^2x(t)}{dt^2}$

(e) $x(3t - 1)$ [for this part, first determine the period of $x(3t - 1)$]

1 3.40

3.40. Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k. Derive the Fourier series coefficients of each of the following signals in terms of a_k:

(a) $x(t-t_0)+x(t+t_0)$

(b) $\mathcal{E}v\{x(t)\}$

(c) $\mathcal{R}e\{x(t)\}$

(d) $\frac{d^2x(t)}{dt^2}$

(e) $x(3t-1)$ [for this part, first determine the period of $x(3t-1)$]

(a)

$$b_k=\frac{1}{T}\int_T x\left(t-t_0\right)e^{-jk(2\pi/T)t}dt$$

$$=\frac{e^{-jk(2\pi/T)t_0}}{T}\int_T x(\tau)e^{-jk(2\pi/T)\tau}d\tau$$

$$=e^{-jk(2\pi/T)t_0}a_k$$

$$c_k=e^{jk(2\pi/T)t_0}a_k.$$

$$d_k=b_k+c_k=e^{-jk(2\pi/T)t_0}a_k+e^{jk(2\pi/T)t_0}a_k=2\cos(k2\pi t_0/T)a_k.$$

1

3.40

3.40. Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k. Derive the Fourier series coefficients of each of the following signals in terms of a_k:

(a) $x(t - t_0) + x(t + t_0)$

(b) $\mathcal{E}v\{x(t)\}$

(c) $\mathcal{R}e\{x(t)\}$

(d) $\frac{d^2 x(t)}{dt^2}$

(e) $x(3t - 1)$ [for this part, first determine the period of $x(3t - 1)$]

(b)

$$\varepsilon v\{x(t)\} = \frac{x(t) + x(-t)}{2}$$

$$b_k = \frac{1}{T}\int_T x(-t)e^{-jk(2\pi/T)t}dt$$

$$= \frac{1}{T}\int_T x(\tau)e^{jk(2\pi/T)\tau}d\tau$$

$$= a_{-k}$$

$$c_k = \frac{a_k + b_k}{2} = \frac{a_k + a_{-k}}{2}$$

1 3.40

3.40. Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k. Derive the Fourier series coefficients of each of the following signals in terms of a_k:

(a) $x(t - t_0) + x(t + t_0)$

(b) $\mathcal{E}v\{x(t)\}$

(c) $\mathcal{R}e\{x(t)\}$

(d) $\frac{d^2x(t)}{dt^2}$

(e) $x(3t - 1)$ [for this part, first determine the period of $x(3t - 1)$]

(c)

$$\mathrm{Re}\{x(t)\} = \frac{x(t) + x^*(t)}{2}$$

$$b_k = \frac{1}{T}\int_T x^*(t)e^{-jk(2\pi/T)t}dt$$

$$b^*{}_k = \frac{1}{T}\int_T x(\tau)e^{jk(2\pi/T)\tau}d\tau = a_{-k}$$

$$b_k = a^*_{-k}$$

$$c_k = \frac{a_k + b_k}{2} = \frac{a_k + a^*_{-k}}{2}$$

1

3.40

3.40. Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k. Derive the Fourier series coefficients of each of the following signals in terms of a_k:

(a) $x(t - t_0) + x(t + t_0)$

(b) $\mathcal{E}v\{x(t)\}$

(c) $\mathcal{R}e\{x(t)\}$

(d) $\frac{d^2 x(t)}{dt^2}$

(e) $x(3t - 1)$ [for this part, first determine the period of $x(3t - 1)$]

(d)

$$\frac{d^2 x(t)}{dt^2} = \sum_{k=-\infty}^{\infty} -k^2 \frac{4\pi^2}{T^2} a_k e^{j(2\pi/T)kt} = \sum_{k=-\infty}^{\infty} b_k e^{j(2\pi/T)kt}$$

$$b_k = -k^2 \frac{4\pi^2}{T^2} a_k$$

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t}$$

$$\frac{dx(t)}{dt} = \frac{d \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_o t}}{dt}$$

$$= \sum_{k=-\infty}^{+\infty} jk\omega_o a_k e^{jk\omega_o t}$$

1 3.40

3.40. Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k. Derive the Fourier series coefficients of each of the following signals in terms of a_k:

(a) $x(t - t_0) + x(t + t_0)$

(b) $\mathcal{E}v\{x(t)\}$

(c) $\mathcal{R}e\{x(t)\}$

(d) $\frac{d^2 x(t)}{dt^2}$

(e) $x(3t - 1)$ [for this part, first determine the period of $x(3t - 1)$]

(e) The period of $x(3t)$ is a third of The period of $x(t)$. Therefore, the signal $x(3t-1)$ is period with period T/3. the FS coefficients of $x(3t)$ are still a_k .Using the analysis of part (a),we know that the FS coefficients of $x(3t-1)$ is $e^{-jk(6\pi/T)}a_k$

1 3.40

3.5.4 Time Scaling

Time scaling is an operation that in general changes the period of the underlying signal. Specifically, if $x(t)$ is periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$, then $x(\alpha t)$, where α is a positive real number, is periodic with period T/α and fundamental frequency $\alpha\omega_0$. Since the time-scaling operation applies directly to each of the harmonic components of $x(t)$, we may easily conclude that the Fourier coefficients for each of those components remain the same. That is, if $x(t)$ has the Fourier series representation in eq. (3.38), then

$$x(\alpha t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(\alpha\omega_0)t}$$

is the Fourier series representation of $x(\alpha t)$. We emphasize that, while the Fourier coefficients have not changed, the Fourier series representation *has* changed because of the change in the fundamental frequency.

1 **3.40**

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$$

$$y(t) = x(at),\ T' = \frac{T}{a},\ \omega' = a\omega$$

$$b_k = \frac{a}{T}\int_{T/a} x(at)e^{-jka\omega_0 t}dt \qquad \qquad Let\ \tau = at,\ d\tau = adt$$

$$= \frac{a}{T}\int_T \frac{1}{a}x(\tau)e^{-jk\omega_0\tau}d\tau$$

$$= \frac{1}{T}\int_T x(\tau)e^{-jk\omega_0\tau}d\tau$$

Answer 3.8

3.8 Since x(t) is real and odd(clue 1), its Fourier series coefficients a_k are purely imaginary and odd (See Table 3.1) Therefore , $a_k = -a_{-k}$ and $a_0 = 0$,Also since it is given that $a_k = 0$ for $|k| > 1$, the only unknown Fourier series coefficients are a_1 and a_{-1} .Using Parseval's relation

$$\frac{1}{T}\int_{<T>} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

for the given signal we have

$$\frac{1}{2}\int_0^2 |x(t)|^2 dt = \sum_{k=-1}^{1} |a_k|^2$$

Using the information given in clue (4) along with the above equation ,

$$|a_1|^2 + |a_{-1}|^2 = 1 \quad \Rightarrow \quad 2|a_1|^2 = 1$$

Therefore

$$a_1 = -a_{-1} = \frac{1}{\sqrt{2}j} \quad \text{or} \quad a_1 = -a_{-1} = -\frac{1}{\sqrt{2}j}$$

The two possible signals which satisfy the given information are

$$x_1(t) = \frac{1}{\sqrt{2}j} e^{j(2\pi/2)t} - \frac{1}{\sqrt{2}j} e^{-j(2\pi/2)t} = -\sqrt{2}\sin(\pi t)$$

and

$$x_2(t) = -\frac{1}{\sqrt{2}j} e^{j(2\pi/2)t} + \frac{1}{\sqrt{2}j} e^{-j(2\pi/2)t} = \sqrt{2}\sin(\pi t)$$

Answer 3.34

3.34.The frequency response of the system is given by

$$H(j\omega)=\int_{-\infty}^{\infty} e^{-4|t|}e^{-j\omega t}dt=\frac{1}{4+j\omega}+\frac{1}{4-j\omega}$$

(a) Here, T=1 and ω_0 =2π and $a_k=1$.for all k. The FS coefficients of the output are

$$b_k=a_kH(j\omega_0k)=\frac{1}{4+j2\pi k}+\frac{1}{4-j2\pi k}$$

(b) Here, T=2 and $\omega_0=\pi$ and,

$$a_k=\begin{cases}0, k even\\ 1, k odd\end{cases}$$

Therefore, the FS coefficients of the outputs are

$$b_k=a_kH(jk\omega_0)=\begin{cases}0, k even\\ \frac{1}{4+jk\pi}+\frac{1}{4-jk\pi}, k odd\end{cases}$$

(c) Here, T = 1,w_0 =2π and

$$a_k=\begin{cases}1/2, & k=0\\ 0, & k \text{ even}, k\neq 0\\ \frac{\sin(\pi k/2)}{\pi k}, & k \text{ odd}\end{cases}$$

Therefore, the FS coefficients of the output are

$$b_k=a_kH(jk\omega_0)=\begin{cases}1/4, & k=0\\ 0, & k \text{ even}, k\neq 0\\ \frac{\sin(\pi k/2)}{\pi k}\left[\frac{1}{4+j2\pi k}+\frac{1}{4-j2\pi k}\right], & k \text{ odd}\end{cases}.$$

Answer 3.40

3.40 Let the FS coefficients of x(t) be a_k

(a)x(t-t_0) is also periodic with period T. The FS coefficients b_k of x(t-t_0) are

$$\begin{aligned} b_k &= \frac{1}{T}\int_T x(t-t_0)e^{-jk(2\pi/T)t}dt \\ &= \frac{e^{-jk(2\pi/T)t_0}}{T}\int_T x(\tau)e^{-jk(2\pi/T)\tau}d\tau \\ &= e^{-jk(2\pi/T)t_0}a_k \end{aligned}$$

Similarly, the FS coefficients of x(t-t_0) are

$$c_k = e^{jk(2\pi/T)t_0}a_k.$$

Finally, the FS coefficients of x(t-t_0)+ x(t+t_0) are

$$d_k = b_k + c_k = e^{-jk(2\pi/T)t_0}a_k + e^{jk(2\pi/T)t_0}a_k = 2\cos(k2\pi t_0/T)a_k$$

Answer 3.40

(b)Note that $\mathcal{E}v\{x(t)\}=[x(t)+x(-t)]/2$. the FS coefficients of x(-t) are

$$\begin{aligned} b_k &= \frac{1}{T}\int_T x(-t)e^{-jk(2\pi/T)t}dt \\ &= \frac{1}{T}\int_T x(\tau)e^{jk(2\pi/T)\tau}d\tau \\ &= a_{-k} \end{aligned}$$

Therefore, the FS coefficients of $\mathcal{E}v\{x(t)\}$ are

$$c_k=\frac{a_k+b_k}{2}=\frac{a_k+a_{-k}}{2}.$$

(c) Note that $\mathrm{Re}\{x(t)\}=[x(t)+x^*(t)]/2$. the FS coefficients of $x^*(t)$ are

$$b_k=\frac{1}{T}\int_T x^*(t)e^{-jk(2\pi/T)t}dt.$$

Conjugating both sides ,we get

$$b_k^*=\frac{1}{T}\int_T x(t)e^{jk(2\pi/T)t}dt=a_{-k}.$$

Therefore, the FS coefficients of $\mathrm{Re}\{x(t)\}$ are

$$c_k=\frac{a_k+b_k}{2}=\frac{a_k+a_{-k}^*}{2}$$

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(d) the FS synthesis equation gives

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j(2\pi/T)kt}.$$

Differentiating both sides wrt t twice ,we get

$$\frac{d^2x(t)}{dt^2} = \sum_{k=-\infty}^{\infty} -k^2\frac{4\pi^2}{T^2}a_k e^{j(2\pi/T)kt}.$$

$-k^2w_0^2a_k$

By inspection, we know that the FS coefficients of $d^2x(t)/dt^2$ are $k\frac{4\pi^2}{T^2}a_k$.

(e) The period of $x(3t)$ is a third of The period of $x(t)$. Therefore, the signal $x(3t-1)$ is period with period T/3. the FS coefficients of $x(3t)$ are still a_k .Using the analysis of part (a),we know that the FS coefficients of $x(3t-1)$ is $e^{-jk(6\pi/T)}a_k$